Emanuel André Medina Araujo

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Education

Technical University of Munich — Oct. 2026 – Present
Master of Science in Informatics — *Munich, DE*

Technical University of Munich — Oct. 2020 – Aug. 2026
Bachelor of Science in Computer Science, Application Area in Physics — *Munich, DE*

Technical University of Munich — Oct. 2024 – Sep. 2025
Parallel studies in Physics (B.Sc. programme, not completed) — *Munich, DE*

Herzog-Christoph-Gymnasium — Sep. 2012 – Sep. 2020
Abitur, Awarded MINT-EC certificate with distinction — *Beilstein, DE*

Experience

Working Student Software Engineer — Nov. 2024 – Present
Vector Informatik GmbH - VSceneCreator (3D OPENDrive Editor) — *Munich, DE*

- Helped ship the 0.1 to 0.4 alpha versions through 43 merged pull requests, working directly with Product Management.
- Built a polyline trajectory editing mode with OpenSCENARIO load and save (Unity/C#): data model, controllers, inspectors, validation issues, and a lossless round-trip of optional XML attributes.
- Designed geometry simplification for OpenDRIVE road networks: detects consecutive line and arc segments with matching heading and curvature and merges them through single or bulk fixes, with undo/redo, live validation and PlayMode tests.
- Cut CI time for editor-only pull requests from 29.8 to 18.8 min and from 43.9 to 33.7 min on two build nodes (Jenkins) by skipping unrelated packaging, reusing module results within a run, and retrying flaky tests in place.
- Built a read-only Explore Mode, the default without a licence: hover readout of world and road (s/t) coordinates, an overview tree of the whole network, and a 1,200-line PlayMode test suite.
- Reduced edit-mode switching delays on large networks (about 2 min for a 40 MB file) by creating editing handles on selection instead of up front; co-developed a staged loading pipeline with global progress.

Working Student Software Engineer — Mar. 2021 – Oct. 2024
Vector Informatik GmbH - DYNA4 (Simulation environment for virtual driving tests) — *Munich, DE*

- Merged 218 pull requests across nine releases (6.0 to 9.1) in Java and Eclipse RCP, working directly with Product Management.
- Engineered a new editor and file type for signal aliases and favourites, with database sync and project migration.
- Built reference analysis so users save only the files a database sync references; co-developed warning markers and a quick fix for simulations with unsaved parameter files, removing a UI-thread deadlock risk from background jobs.
- Authored 50+ test pull requests (SWTBot, JUnit), including reusable wrappers for an in-house form framework and a JUnit formatter that feeds CI results into test management (Xray).

Projects

LoveNeverDies | *Legal Loves Tech Hackathon – Bavarian Ministry of Justice Challenge* — Aug. 2026

- Built an end-to-end encrypted estate assistant for bereaved families (React, TypeScript, Supabase) in ten days with a team of lawyers, guiding users through inheritance deadlines via a 141-node legal decision tree.
- Designed a zero-trust architecture in which the server only stores ciphertext: hybrid post-quantum key exchange and signatures (ML-KEM + X25519, ML-DSA + Ed25519), AES-256-GCM, and a Shamir-secret-shared vault.
- Verified row-level security against real Postgres migrations with 130 Vitest files and multi-user Playwright E2E tests.

Learning Based Inverse Kinematics | *Bachelor Thesis* — Jun. 2024 – Nov. 2024

- Compared learning-based IK solvers (MLP, LSTM, hybrid, PPO) for 2- and 3-DOF manipulators in PyTorch on a 10k-sample test set; a simple MLP with direct-angle prediction consistently outperformed the more complex models.

Ironman | *3.8km swim, 180km bike, 42.2km run; Overall Time : 14:21:26h* — Oct. 2022 – Aug. 2024

- Completed a full Ironman triathlon through disciplined training and a goal-oriented race plan.

Technical Skills

Languages: C#, Java, TypeScript, JavaScript, Python, C/C++, SQL, MATLAB
Frameworks: React, React Native, Unity, PyTorch, Eclipse RCP, EMF, JavaFX, Supabase/PostgreSQL, WebRTC
Developer Skills: Git, Linux, GitHub Actions, Jenkins, Conan, SWTBot, JUnit, Playwright, Vitest, Jira, OpenDRIVE, OpenSCENARIO